

A “Mathematical” Tutorial on Semi- and nonparametric Efficient Estimation Theory

PLSC 40601 (Spring 2026) Advanced Topics in Causal Inference *

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1 Introduction and Goals

The statistical science community has gradually moved away from relying solely on fully parametrized statistical models. Traditional estimation and inference tools based on fully parametric models have been largely outdated. For instance, there is little justification to start with the belief that the data-generating process (DGP) is perfectly specified by a linear model and follows a Gaussian distribution $N(\mu, \Sigma)$. Instead, researchers are increasingly adopting semiparametric, or simply nonparametric, approaches in estimating and conducting causal inference on treatment and structural parameters of interest. Relying on much weaker assumptions about the true world, these new methods also have huge potential when integrated with machine learning (ML) tools, allowing for much more flexible estimators while preserving ideal asymptotic properties and robustness.

However, it has come to my attention that the required mathematical technicalities constituted a pedagogical barrier for the broader social science community. It is not uncommon to see the complicated mathematics and abstract notations impedes students' reading and understanding of the literature in the field of semi-and nonparametric theory. This project serves as a “mathematical” tutorial oriented to a broader audience in the quantitative social science disciplines, aiming to provide a condensed reading guide to the classic theory in semi-and nonparametric estimation.

The following pages are divided as the following. A critical contribution of this project is Section 2, which features a collection of advanced mathematical tools and concepts frequently used in the semi-parametric theory literature. It mainly focuses on three areas, including functional analysis, likelihood and Fisher information, and a quick re-formulation of core concepts in probability theory via measure-theoretic language, which can be beneficial for understanding proofs. Section 3 is largely based on the review paper by [Kennedy \(2024\)](#).

After a short recapture of parametric model, efficient estimator, and Cramér-Rao lower bound, we focus on the central idea of using submodels to derive the nonparametric equivalence of Cramér-Rao lower bound. Critical concepts like pathwise differentiability and influence function will be introduced along the way.

Section 4 draws heavily from the work by Chernozhukov et al. (2018) to introduce the powerful concept of Neyman orthogonality score, show how to construct orthogonal score function, and illustrate the usefulness of orthogonality via a worked out solution for estimating the Average Treatment Effect (ATE) using doubly-debiased estimator.

The main text of this tutorial focuses on building mathematical intuition and providing descriptive interpretation of abstract mathematical concepts. Although theorems and definitions are going to be formally stated, proofs and mathematical rigor are of secondary considerations. The Appendix collects proofs of some important results and mathematical facts. Mathematically-oriented readers are encouraged to use it as a reference.

2 Preliminary Mathematics and Notations

2.1 Functional Analysis

Many estimands of interest—usually some sort of conditional expectation functions—often take in a function as an input and evaluates to a real number. To facilitate the discussion of semi-parametric estimation and Neyman orthogonality, we first provide a crash course on functionals and functional derivatives.

Definition 1. A functional defined most generally is a mapping:

$$\Psi : \mathcal{X} \rightarrow \mathcal{Y},$$

where $\mathcal{X}$ is a normed vector space (usually a Hilbert space in our context) and $\mathcal{Y}$ is commonly real-valued vector space (most often $\mathbb{R}$ or $\mathbb{R}^d$)

In our context, it is best to think in the case of $\Psi : \mathcal{P} \rightarrow \mathbb{R}$, where each element in $P \in \mathcal{P}$ is a distribution function. The collection $\mathcal{P}$ is referred to as a model space. See Table 1 for some common mathematical objects in causal inference that are examples of functionals.

Gâteaux And Fréchet Derivatives: Given a functional $\Psi : \mathcal{P} \rightarrow \mathbb{R}$, the Gâteaux derivative at P in direction h is defined as

$$D\Psi(P)[h] = \lim_{\epsilon \rightarrow 0} \frac{\Psi(P + \epsilon h) - \Psi(P)}{\epsilon}. \quad (1)$$

Estimands	Functional Expressions
Mean functional	$\Psi(P) = \mathbb{E}_P[X] = \int x dP(x)$
General expectation functional	$\Psi(P) = \mathbb{E}_P[g(X)] = \int g(x) dP(x)$
Variance functional	$\Psi(P) = \mathbb{E}_P[X^2] - (\mathbb{E}_P[X])^2$
Conditional expectation functional	$\Psi(P)(x) = \mathbb{E}_P[Y \mid X = x]$
Average Treatment Effect (ATE)	$\Psi(P) = \mathbb{E}_P[Y(1) - Y(0)]$
Entropy functional	$\Psi(P) = - \int p(x) \log p(x) dx$

Table 1: Examples of functionals.

This is reminiscent of the definition of derivative in real-valued functions. A small perturbation in the direction of a function h can result in a small perturbation of the functional. Varying the value of ϵ results in model paths that traverses through the model space to form submodels. We often can denote : $P_\epsilon = P + \epsilon h, \epsilon \in \mathbb{R}$, and write:

$$D\Psi(P)[h] = \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} \Psi(P_\epsilon),$$

because the tangency generates the path. To assist with understanding, one can think of it as a generalized directional derivative in multivariate calculus. Suppose $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, the directional derivative along direction of $u = (a, b)'$ is defined as:

$$D(f)[u] = \nabla f(x, y) \cdot \frac{u}{\|u\|}$$

If $\mathcal{P}$ is normed (often Banach) space, we say Ψ is Fréchet differentiable at P if there exists

a bounded operator $D\Psi(P)[h]$ such that

$$\lim_{\|h\| \rightarrow 0} \frac{\Psi(P+h) - \Psi(P) - D\Psi(P)[h]}{\|h\|} = 0.$$

Heuristically, one can think of it as the first-order, tangent plane approximation to a real-valued surface. Gateaux differentiability is a pathwise statement, as the derivative has to exist only along a direction, but the Frechet differentiability requires

$$\Psi(P+h) = \Psi(P) + D\Psi(P)[h] + o(\|h\|).$$

Similarly, one can find intuition for Frechet differentiability in the real-valued function analog. Consider the usual function we deal with $f : \mathbb{R} \rightarrow \mathbb{R}$. We say f is differentiable at $x = a$ if the following limit exists and the derivative at a is the limit value.

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = f'(a) \quad (2)$$

Rewriting (2) in the following way reveals that the derivative provides simply the best **affine approximation** of f at a :

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a) - hf'(a)}{h} = 0, \quad \Delta f \approx f(a+h) - f(a) = hf'(a), \quad (3)$$

and notice the Gateaux derivative $D\Psi(P)[h]$ in (1) plays a similar role of approximating the functional Ψ with a linear functional.

2.2 Measure-based Probability Theory

Definition 2 (Probability Space). Given a generic sample space Ω , we endow it with a σ -algebra $\mathcal{F}$. Define a probability measure $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$. We call the triple $(\Omega, \mathcal{F}, \mathbb{P})$ a probability space. A measure space is a more general triple of the form $(X, \mathcal{S}, \mu)$.

Definition 3 (Random Variables, Expectation, and Variance). Given a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, a random variable X is an $\mathcal{F}$ -measurable function: $X : \Omega \rightarrow \mathbb{R}$. A realization of the random variable is $X(\omega) = x \in \mathbb{R}$, $\omega \in \Omega$. The expectation and variance of X are defined as the Lebesgue integrals:

$$\mathbb{E}_{\mathbb{P}}[X] = \int_{\Omega} X d\mathbb{P}(\omega), \quad Var[X] = \mathbb{E}[(X - \mathbb{E}[X])^2]. \quad (4)$$

We note that if there exists a density function $p(x)$ for $\mathbb{P}$ such that $d\mathbb{P} = p$, the expectation

can be equivalently characterized by the usual Riemann integral:

$$\mathbb{E}[X] = \int_{\mathbb{R}} x p(x) d\lambda(x),$$

where $\lambda(\cdot)$ is the Lebesgue measure on the real numbers.

As suggested above, we are dealing with functionals, so it is essential that we define a normed vector space. Consider the $L^q(\mathbb{P})$ norm below.

Definition 4 ($L^q(\mathbb{P})$ Space). Given any probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and take $q \in \mathbb{Z}^+$, we define:

$$L^q(\mathbb{P}) = \{f : \Omega \rightarrow \mathbb{R} \text{ is } \mathcal{F} - \text{measurable} : \|f\|_q \leq \infty\}, \quad (5)$$

where the norm $\|\cdot\|_q$ is defined as

$$\|f\|_q = \left(\int |f|^q d\mathbb{P} \right)^{1/q}. \quad (6)$$

Definition 5 (Inner Product and Cauchy-Schwartz Inequality of Random Variables). Given $L^2(\mathbb{P})$, where the 2-norm is defined as (6). We recognize that it is a Hilbert Space with inner product: for two random variables $X, Y \in L^2(\mathbb{P})$,

$$\langle X, Y \rangle = \int_{\Omega} X \cdot Y d\mathbb{P}(\omega) = \mathbb{E}_{\mathbb{P}}[XY]. \quad (7)$$

Realize that the 2-norm of a random variable is induced by its inner product with itself:

$$\sqrt{\langle X, X \rangle} = \left(\int_{\Omega} X^2 d\mathbb{P}(\omega) \right)^{1/2} = \sqrt{\mathbb{E}_{\mathbb{P}}[X^2]}, \quad \sqrt{\langle Y, Y \rangle} = \left(\int_{\Omega} Y^2 d\mathbb{P}(\omega) \right)^{1/2} = \sqrt{\mathbb{E}_{\mathbb{P}}[Y^2]}. \quad (8)$$

Combining (7) and (8), we have the random-variable version of the **Cauchy-Schwartz Inequality**:

$$|\langle X, Y \rangle| \leq \|X\|_2 \cdot \|Y\|_2 \implies \mathbb{E}_{\mathbb{P}}[XY]^2 \leq \mathbb{E}_{\mathbb{P}}[X^2] \cdot \mathbb{E}_{\mathbb{P}}[Y^2] \quad (9)$$

2.3 Likelihood Theory and Fisher Information

Lastly, since many score functions attain the form of log-likelihood or score function, we specify a few definitions and two useful theorems related to likelihood theory. In this subsection, let $z_1, z_2, \dots, z_n \stackrel{i.i.d}{\sim} P_{\theta}$, where $\{P_{\theta} : \theta \in \Theta \subset \mathbb{R}^d\}$ is a model space with well-defined probability densities $\{p_{\theta}(z_i)\}_{i=1}^n$. As we know the likelihood of a single observation is just the density $p_{\theta}(z_i)$ itself and the log-likelihood is

$$\ell(\theta) = \log p_{\theta}(z_i). \quad (10)$$

For the full sample, the likelihood and log-likelihood functions are

$$L_n(\theta; Z) = \prod_{i=1}^n p_\theta(z_i), \quad \ell_n(\theta) = \sum_{i=1}^n \log p_\theta(z_i). \quad (11)$$

Definition 6 (Score Function). We define the individual score function as the following:

$$S_\theta(z) = \nabla_\theta \ell(\theta) \quad (12)$$

Two important results about the score function are worth noting. Theorem 1 below says score function as zero mean, and Theorem 2 motivates the definition of the Fisher Information. The full proofs of these two results can be found in Appendix A.

Theorem 1 (Score function has mean zero).

$$\mathbb{E}[\nabla_\theta \ell(\theta)] = 0. \quad (13)$$

Theorem 2 (Information Equality under correct model specification). *Under correct model specification, meaning the parametric family $\{P_\theta : \theta \in \Theta \subset \mathbb{R}^d\}$ captures the true parameter θ_0 , we have the following equality:*

$$-\mathbb{E}[\nabla_\theta^2 \ell(\theta)] = \text{Var}[\nabla_\theta \ell(\theta)]. \quad (14)$$

Definition 7 (Fisher Information Matrix). Fisher Information is a matrix $I(\theta)$ that is equal to the Hessian of the log-likelihood function. Specifically, the matrix has entry-wise element:

$$[I(\theta)]_{i,j} = \left[\frac{\partial^2}{\partial \theta_i \partial \theta_j} \ell(\theta) \right]. \quad (15)$$

Fisher information is a crucial concept in the asymptotics of nonlinear model, typically estimated via the Maximum Likelihood approach. The Maximum Likelihood Estimator (MLE) defined as the solution to the optimization problem:

$$\hat{\theta}_n \in \arg \max_{\theta \in \Theta} L_n(\theta) = \arg \max_{\theta \in \Theta} \ell_n(\theta),$$

and under regularity conditions behaves well asymptotically:

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} \mathcal{N}(0, I(\theta_0)^{-1}),$$

where we observe the asymptotic variance of the MLE estimator is exactly the inverse of Fisher Information. Even better, the MLE satisfies the parametric efficiency bound, meaning

$I^{-1}(\theta_0)$ is the lowest variance achievable—an idea generalized into the following theorem.

Theorem 3 (Cramér–Rao Lower Bound (CRLB)). *Let $\{P_\theta : \theta \in \Theta \subset \mathbb{R}\}$ be a parametric model with density $p_\theta(z)$ satisfying standard regularity conditions. Let $T = T(Z_1, \dots, Z_n)$ be an unbiased estimator of θ , i.e.,*

$$\mathbb{E}_\theta[T] = \theta.$$

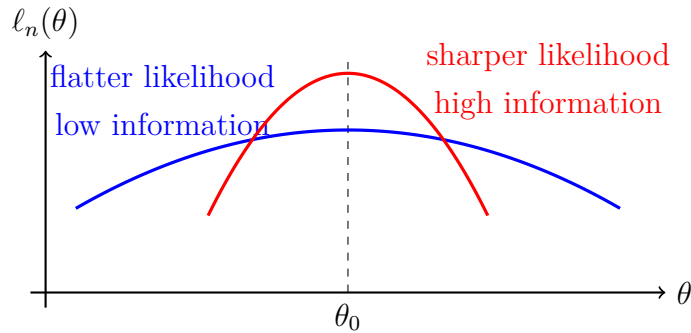
Then the variance of T satisfies:

$$\text{Var}_\theta(T) \geq \frac{1}{I_n(\theta)},$$

where $I_n(\theta)$ is the Fisher information in the sample:

$$I_n(\theta) = n \cdot I(\theta).$$

Visually, consider a uni-dimensional parameter setting. More information means the likelihood is more sharply curved around the truth.



2.4 Common Notations

Going forward, we stipulate some conventions and shorthand notations.

- Greek letters are reserved for parameters of interest, unless specified otherwise
- A subscript of 0 means the true parameter value
- P denotes a generic probability measure, and $\mathbb{P}$ usually denotes the “true” probability function
- $\Pr, \mathbb{E}_P$ denotes the probability and expectation operator with respect to measure P
- $W = w$ or $Z = x$ denote random variables and their observed realizations

- $\|\cdot\|_{P,q}$ denotes the $L^q(P)$ norm:

$$\left(\int |f(w)|^q dP(w) \right)^{1/q}$$

- For a function f differentiable in $x \in \mathbb{R}^d$, we use:

$$\partial_{x'} f = \frac{\partial}{\partial x'} f, \quad \partial_{x'} f(x_0) = \frac{\partial}{\partial x'} f|_{x=x_0}$$

3 From Parametric to Nonparametric Estimation

3.1 A Quick Recap of Smooth Parametric Model Estimation

In the parametric context, our model space is a collection $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$. For demonstrative purposes, we consider $\Theta \subset \mathbb{R}$. Suppose we have a target to learn, expressed as a functional $\psi(\theta)$. We estimate $\hat{\psi}$ from data Z . The core issue this section deals with is finding a benchmark for the efficiency of different estimators $\hat{\psi}$, defined in terms of Mean Squared Error (MSE) loss. A classical result for smooth parametric models is the Crámer–Rao lower bound.

Suppose that both P_θ and $\psi(\theta)$ are differentiable in θ . Then for an unbiased estimator $\hat{\psi}$ of $\psi(\theta)$,

$$\text{Var}(\hat{\psi}) \geq \frac{(\psi'(\theta))^2}{\text{Var}(S_\theta(Z))}, \quad (16)$$

where

$$\psi'(\theta) = \frac{\partial}{\partial \theta} \psi(\theta) \quad \text{and} \quad S_\theta(z) = \frac{\partial}{\partial \theta} \log p_\theta(z) \quad (17)$$

is the score function. In this expression, $\text{Var}(S_\theta(Z))$ is the Fisher information, and the bound identifies the smallest achievable variance among unbiased estimators of $\psi(\theta)$.

More generally, one can benchmark an estimator through a common trick of minimax lower bound (Kennedy, 2024, 6):

$$\inf_{\hat{\psi}} \sup_{P \in \mathcal{P}} \mathbb{E}_P \left[(\hat{\psi} - \psi(P))^2 \right] \geq R_n.$$

Taking the first layer of supremum takes the worst-case mean squared error over the entire model class $\mathcal{P}$, then taking the infimum select one estimator out of all candidate estimators that minimizes the worst-case MSE.

A natural question to ask, then, is can the idea of Cramér–Rao lower bound (16) and

parametric efficiency be generalized to the nonparametric setting?

3.2 Parametric Submodels

Now we are in the case where the model space is much more flexible. Namely, $\mathcal{P}$ can contain any well-defined probability distribution function P , which makes it more challenging to find the target $\psi(\mathbb{P})$, where $\mathbb{P}$ denotes the true probability measure, since we have to travel through a much larger space. How do we connect the Cramér–Rao lower bound to semi- or non-parametric models? The key is to invoke the idea *parametric submodels*.

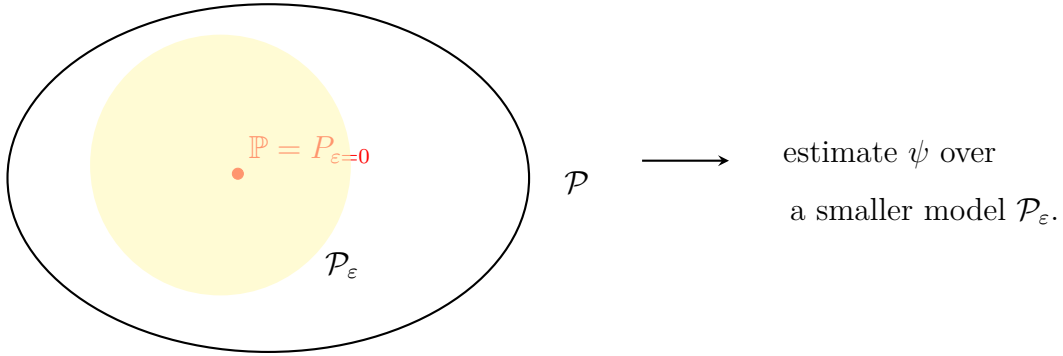
Smooth submodels Let the full model be $\mathcal{P}$, and consider a smooth one-dimensional submodel

$$\mathcal{P}_\varepsilon = \{P_\varepsilon : \varepsilon \in \mathbb{R}\}$$

such that

$$\mathcal{P}_\varepsilon \subseteq \mathcal{P}, \quad P_{\varepsilon=0} = P. \quad (18)$$

What 18 means is that the submodels is contained in the overarching model space and hence is easier to travel through. Also, when evaluating ε at zero, we recover the true probability distribution. Then we estimate the target $\psi(\mathbb{P})$ by studying it along the smaller model $\mathcal{P}_\varepsilon$.



Non-parametric model via a submodel For a non-parametric model $\mathcal{P}$, we usually construct a one-parameter submodel using:

$$dP_\varepsilon = d\mathbb{P}(z)\{1 + \varepsilon h(z)\}, \quad (19)$$

where $h : Z \rightarrow \mathbb{R}$ is a mean-zero function satisfying

$$\|h\|_\infty < \infty, \quad |\varepsilon| \text{ small enough so that } p_\varepsilon(z) \geq 0.$$

Construction (19) represents a local h -perturbation away from a nonparametric model $P \in \mathcal{P}$, controlled by ε . What's more, we notice that the corresponding density p_ε of the perturbed probability distribution P_ε has the following nice property:

$$\left. \frac{\partial}{\partial \varepsilon} \log p_\varepsilon(z) \right|_{\varepsilon=0} = \left. \frac{\partial}{\partial \varepsilon} \log(1 + \varepsilon h(z)) \right|_{\varepsilon=0} = \left. \frac{h(z)}{1 + \varepsilon h(z)} \right|_{\varepsilon=0} = h(z), \quad (20)$$

meaning the score for this submodel at $\varepsilon = 0$ is exactly $h(z)$.

To examine how local perturbation of probability measure impact the learning of $\psi(P)$, we invoke the following distributional analog of Taylor expansion.

Definition 8 (von Mises expansion). For a smooth functional $\psi : \mathcal{P} \rightarrow \mathbb{R}$, define the *distributional Taylor expansion* by

$$\psi(\bar{P}) - \psi(P) = \int \varphi(z; \bar{P}) d(\bar{P} - P)(z) + R_2(\bar{P}, P), \quad (21)$$

for $\bar{P}, P \in \mathcal{P}$, where

$$\int \varphi(z; P) dP(z) = 0, \quad \int \varphi(z; P)^2 dP(z) < \infty.$$

The smoothness of the submodel and expansion (21) give rise to the following crucial result in semi- and nonparametric efficient estimation theory: **pathwise differentiability**.

A functional $\psi : \mathcal{P} \rightarrow \mathbb{R}$ is said to be *pathwise differentiable* at P if for every regular parametric submodel P_ε

$$\left. \frac{d}{d\varepsilon} \psi(P_\varepsilon) \right|_{\varepsilon=0} = \int \varphi(z; \mathbb{P}) S_\varepsilon(z) d\mathbb{P}(z), \quad (22)$$

where the function φ is often called the **influence function**, as it quantifies exactly how the local perturbation via the submodel influences the target. A full proof to statement (22) can be found in Appendix A.

3.3 Pathwise Differentiability and Generalization of Cramér–Rao bound

Here, we show that the detour we went through above can be used to generalize the Cramér–Rao bound, via pathwise differentiability.

First observe that the Cramér–Rao lower bound for a given $P_\varepsilon \in \mathcal{P}_\varepsilon$ is

$$\frac{\psi'(P_\varepsilon)^2}{\text{Var}_{P_\varepsilon}(S_\varepsilon(Z))} = \frac{\left(\frac{\partial}{\partial \varepsilon} \psi(P_\varepsilon)\right)^2 \Big|_{\varepsilon=0}}{\mathbb{E}[h(Z)^2]} = \frac{\mathbb{E}[\varphi(z, \mathbb{P})h(z)]^2}{\mathbb{E}[h(Z)^2]},$$

where the first equality follows from definition and *i.i.d* assumption, and the second equality follows from (22), specifically

$$\int \varphi(z; \mathbb{P}) S_\varepsilon(z) d\mathbb{P}(z) = \mathbb{E}[\varphi(z, \mathbb{P})h(z)].$$

Then over all the Cramér–Rao bound at $\varepsilon = 0$, we simply take the supremum of all bounds:

$$\sup_{P_\varepsilon} \frac{\psi'(P_\varepsilon)^2}{\text{Var}_{P_\varepsilon}(S_\varepsilon(Z))} = \sup_h \frac{\mathbb{E}[\varphi(z, \mathbb{P})h(z)]^2}{\mathbb{E}[h(z)^2]}. \quad (23)$$

Notice, by the Cauchy-Schwarz Inequality (9), for each h , we have:

$$\mathbb{E}[\varphi(z, \mathbb{P})h(z)]^2 \leq \mathbb{E}[\varphi(z, \mathbb{P})^2] \cdot \mathbb{E}[h(z)^2], \quad (24)$$

which implies $\forall h$,

$$\frac{\mathbb{E}[\varphi(z, \mathbb{P})h(z)]^2}{\mathbb{E}[h(z)^2]} \leq \mathbb{E}[\varphi(z, \mathbb{P})^2] = \text{Var}[\varphi(z)].$$

In another word, we have an upper-bound for $\frac{\mathbb{E}[\varphi(z, \mathbb{P})h(z)]^2}{\mathbb{E}[h(z)^2]}$. Since supremum is the least upper-bound, then (23) becomes

$$\sup_{P_\varepsilon} \frac{\psi'(P_\varepsilon)^2}{\text{Var}_{P_\varepsilon}(S_\varepsilon(Z))} = \sup_h \frac{\mathbb{E}[\varphi(z, \mathbb{P})h(z)]^2}{\mathbb{E}[h(z)^2]} \leq \text{Var}[\varphi(z)] \quad (25)$$

Finally, we now can assert that for an estimator $\hat{\psi}$, it satisfies the **nonparametric efficiency bound** if

$$\sqrt{n}(\hat{\psi} - \psi) \xrightarrow{d} N(0, \text{Var}[\varphi(z)]).$$

4 Neyman Orthogonality and Double/Debiased Machine Learning Estimator

So far we have looked at fully parametrized models and nonparametric models, which are two extreme cases. In the former, we as researchers have a perfectly informative prior that the state of the world works exactly according to our model specification, while in the latter

we do not impose any prior beliefs and assumptions about the data-generating process. With the popularization of machine learning tools, nonparametric inference also has become a conventional tool when the researchers are not sure how control variables should enter the model, which allows flexible model forms (e.g., the regression function $Y = g(X) + U$ does not necessarily have $g(X) = X'\beta$).

Drawing from [Chernozhukov et al. \(2018\)](#), this last chapter presents the intermediate approach: semiparametric estimation. We are interested in the setting where our goal is to estimate a low-dimensional parameter θ_0 , but we are dealing with a high-dimensional nuisance η_0 (e.g., regression adjustment or the propensity score).

The key challenge here is that ML methods applied to learn η_0 is to use regularization bias to trade off high variance in estimator $\hat{\theta}_0$ of θ_0 , but the bias in learning the nuisance parameters also induces bias in $\hat{\theta}_0$, which implies we have a situation where the estimator is not centered at θ_0 and not $\sqrt{n}$ consistent:

$$\sqrt{n}(\hat{\theta}_0 - \theta_0) \xrightarrow{p} \infty. \quad (26)$$

The solutions are two simple but powerful fixes: 1. Neyman Orthogonal Scores and 2. Cross-fitting. The resulting estimators resulting from those procedures are known as Double/Debiased ML estimators, which are unbiased and enjoy the $\sqrt{n}$ consistency.

4.1 Definition and Set-up of Neyman Orthogonality

Neyman Orthogonality Condition We set up the Neyman Orthogonality condition. We have a $\theta_0 \in \Theta \neq \emptyset$, $\Theta \subset \mathbb{R}^{d_\theta}$. The true parameter θ_0 satisfies the moment condition

$$\mathbb{E}_P[\psi(W; \theta_0, \eta_0)] = 0, \quad (27)$$

where $\psi = (\psi_1, \psi_2, \dots, \psi_{d_\theta})'$ is the vector of the score functions, W is the random variable in the probability space $(\mathcal{W}, \mathcal{A}_{\mathcal{W}}, P)$, $P \in \mathcal{P}_N$ is a generic probability measure, η_0 is the true value of **nuisance parameter** $\eta \in T$, T is a convex subset of some normed vector space endowed with norm $\|\cdot\|_T$.

We further assume each score function $\psi_j : \mathcal{W} \times \Theta \times T \rightarrow \mathbb{R}$ is also measurable once we endow a Borel σ -field on Θ and T respectively. Let a sample of observations $(W_i)_{i=1}^N$ be given.

Denote the shifted set of nuisance realizations $\tilde{T} = \{\eta - \eta_0 : \eta \in T\}$, we define the **pathwise (Gâteaux) derivative** as a mapping: $D_r : \tilde{T} \rightarrow \mathbb{R}^{d_\theta}$

$$D_r[\eta - \eta_0] = \partial_r\{\mathbb{E}_P[\psi(W; \theta_0, \eta_0 + r(\eta - \eta_0))], \eta \in T\}, \quad (28)$$

for all $r \in [0, 1)$. Note this r introduced is similar to the idea of model path. Also note that $\psi(W; \theta_0, \eta_0 + r(\eta - \eta_0))$ is well-defined because $\eta_0 + r(\eta - \eta_0) = (1 - r)\eta_0 - r\eta$ is a convex combination of two elements in the convex set T , so $\eta_0 + r(\eta - \eta_0) \in T$.

For convenience, instead of using the path induced by r , we define:

$$\partial_\eta \mathbb{E}_P[\psi(W; \theta_0, \eta_0)][\eta - \eta_0] := D_0[\eta - \eta_0], \quad \eta \in T \quad (29)$$

Denote $\mathcal{T}_N \subset T$ as a set of nuisance realizations such that the estimators $\hat{\eta}_0$ of η specified below take values in this set with high probability. In practice, we typically assume $\mathcal{T}_N$ is a properly shrinking neighborhood of T . The **Neyman Orthogonality Condition** is essentially the derivative (29) vanishes for all $\eta \in \mathcal{T}_N$.

Definition 9 (Neyman Orthogonality). The score vector $\psi = (\psi_1, \dots, \psi_{d_\theta})'$ obeys the orthogonality condition at (θ_0, η_0) with respect to $\mathcal{T}_N \subset T$ if moment condition (27) holds and the pathwise derivative $D_r[\eta - \eta_0]$ exists for all $r \in [0, 1)$ and all $\eta \in \mathcal{T}_N$ and vanishes at $r = 0$, namely

$$\partial_\eta \mathbb{E}_P[\psi(W; \theta_0, \eta_0)][\eta - \eta_0] = 0, \forall \eta \in \mathcal{T}_N \quad (30)$$

We remark that the condition (30) will be satisfied with $\mathcal{T}_N = T$ when η is finite dimensional as long as

$$\partial_\eta \mathbb{E}_P[\psi_j(W; \theta_0, \eta_0)] = 0, \forall j = 1, 2, \dots, d_\theta,$$

where $\partial_\eta \mathbb{E}_P[\psi_j(W; \theta_0, \eta_0)]$ denotes the vector of partial derivatives of the function $\eta \mapsto \mathbb{E}_P[\psi_j(W; \theta_0, \eta)]$ evaluated at $\eta = \eta_0$.

4.2 Construction of Orthogonal Score

Section 4.1 can feel too technical and irrelevant to research practice, so this section first provides motivation for using Neyman Orthogonal score to learn the target parameter and constructing one if the original natural score is not orthogonal. Second, we describe the process of orthogonalizing a score function using a toy example of log-likelihood function and ties it back to the idea that using doubly robust estimator is considered as a superior approach in causal inference.

4.2.1 Motivation: Why orthogonality?

Recall that in semiparametric inference, we have to start from some identification assumption, which is often formed as some sort of GMM-style moment condition:

$$\mathbb{E}[\psi(W; \theta_0, \eta_0)] = 0,$$

where θ is the target parameter and η is the nuisance function. Since it is the population statement, with data $\{W_i\}_{i=1}^n$, we solve the moment condition in its sample analog. Then the estimator $\hat{\theta}$ is the solution to

$$\frac{1}{n} \sum_{i=1}^n \psi(W_i; \hat{\theta}, \hat{\eta}) = 0, \quad (31)$$

where $\hat{\eta}$ is learned from data either parametrically or flexibly using machine learning. The core challenge we are dealing with, to reiterate, is that the regularization bias induced by estimating $\hat{\eta}$ propagates in the asymptotic distribution of $\hat{\theta}$. To see this, consider the Taylor expansion of sample moment condition (31) around the truth (θ_0, η_0) :

$$\begin{aligned} 0 &= \frac{1}{n} \sum_{i=1}^n \psi(W_i; \theta_0, \eta_0) + \underbrace{\frac{1}{n} \sum_{i=1}^n \partial_{\theta} \psi(W_i; \theta_0, \eta_0) (\hat{\theta} - \theta_0)}_{:-J_n} + \\ &\quad \frac{1}{n} \sum_{i=1}^n \partial_{\eta} \psi(W_i; \theta_0, \eta_0) (\hat{\eta} - \eta_0) + \underbrace{o(\|\hat{\eta} - \eta_0\|^2)}_{\text{higher order terms}} \\ \implies -J_n(\hat{\theta} - \theta_0) &= \frac{1}{n} \sum_{i=1}^n \psi(W_i; \theta_0, \eta_0) + \frac{1}{n} \sum_{i=1}^n \partial_{\eta} \psi(W_i; \theta_0, \eta_0) (\hat{\eta} - \eta_0) + o(\|\hat{\eta} - \eta_0\|^2) \\ \text{solve for: } (\hat{\theta} - \theta_0) &= -J_n^{-1} \left(\frac{1}{n} \sum_{i=1}^n \psi(W_i; \theta_0, \eta_0) + \frac{1}{n} \sum_{i=1}^n \partial_{\eta} \psi(W_i; \theta_0, \eta_0) (\hat{\eta} - \eta_0) + o(\|\hat{\eta} - \eta_0\|^2) \right) \end{aligned}$$

Scaling the last line of result above by $\sqrt{n}$ and examining the asymptotic behavior of $\sqrt{n}(\hat{\theta} - \theta_0)$, we can quickly see it does not enjoy nice asymptotic properties:

$$\sqrt{n}(\hat{\theta} - \theta_0) = -J_n^{-1} \left(\underbrace{\frac{1}{\sqrt{n}} \sum_{i=1}^n \psi(W_i; \theta_0, \eta_0)}_{CLT} + \underbrace{\frac{1}{\sqrt{n}} \sum_{i=1}^n \partial_{\eta} \psi(W_i; \theta_0, \eta_0) (\hat{\eta} - \eta_0) + \sqrt{n} o(\|\hat{\eta} - \eta_0\|^2)}_{\text{Trouble Term}} \right), \quad (32)$$

where the first CLT labeled term enjoys a Central Limit Theorem since it is just a sample

average, implied by the law of large numbers and continuous mapping theorem, and the higher order term can be neglected under some structural assumptions (e.g., sparsity or Lasso penalty). The Trouble Term in (32) is problematic due to the following two reasons.

Regularization Bias: neither $\partial_\eta \psi(W_i; \theta_0, \eta_0)$ nor $\hat{\eta} - \eta_0$ has mean zero in general, which means this whole term will not be properly centered asymptotically.

Overfitting Bias: we cannot assume $\partial_\eta \psi(W_i; \theta_0, \eta_0) \perp \hat{\eta} - \eta_0$ because $\hat{\eta}$ is learned from W_i 's, so we cannot split $\partial_\eta \psi$ and $\hat{\eta} - \eta_0$ in expectation operator. More precisely, even if each individual term $\partial_\eta \psi$ and $\hat{\eta} - \eta_0$ is mean zero, we still might have $\mathbb{E}[\partial_\eta \psi(\hat{\eta} - \eta_0)] \neq 0$. This is where the Neyman Orthogonality in Section 4.1 comes to rescue.

Neyman Orthogonal Scores kills regularization bias: Observe that imposing the Neyman Orthogonality condition in (30) mechanically forces the $\partial_\eta \psi(W_i; \theta_0, \eta_0)$ term in (32) to be zero mean. In this view, (30) simply means that the first-order influence of bias in learning the nuisance parameter $\hat{\eta}$ does not propagate to the learning of the distribution of $\hat{\theta}$. We ignore the presence of second and higher-order influence of regularization bias.

Cross-fitting addresses overfitting bias: We split the data $\{W_i\}_{i=1}^n$ into K folds and using parts of the data to learn $\hat{\eta}$ and then use other parts to learn $\psi(W_i; \hat{\theta}, \hat{\eta})$, which gives independence.

The second ingredient can be implemented via off-the-shelf ML methods, and the next order of business is to make sure we can actually find an orthogonal score function ψ .

4.2.2 General Construction of Neyman Orthogonal Scores

Section 4.2.1 motivates the use of an orthogonal score, but it is likely that the original score function that the target parameter of interest admits is NOT orthogonal. Section 4.2.2 offers some reassurance that one can generally construct an orthogonal score based on the original score function for most of the causal parameters common in social scientific research.

Let an original score φ be given, and it does not satisfy the orthogonality condition above, our goal is to *orthogonalize* it into ψ . To explain this better, we discuss the *Neyman orthogonal scores for likelihood and finite-dimensional nuisance parameters*. For other approaches, such as the centering-out, see Chernozhukov et al. (2018).

We play with a toy example by taking $\ell(W; \theta, \beta)$ to be a criterion function, typically a log-likelihood or quasi-log-likelihood, where $\theta \in \Theta \subset \mathbb{R}^{d_\theta}$, for a convex set $\mathcal{B}$, is the target

parameter and $\beta \in \mathcal{B} \subset \mathbb{R}^{d_\beta}$ is the nuisance parameter. The true parameters (θ_0, β_0) solves the familiar optimization question:

$$\max_{\theta \in \Theta, \beta \in \mathcal{B}} \mathbb{E}[l(W; \theta, \beta)] \quad (33)$$

The original score for θ is defined literally as the score function for log-likelihood as (12) Section 2:

$$\varphi(W; \theta, \beta) = \partial_\theta \ell(W; \theta, \beta).$$

Notice (33) implies the moment conditions at the truth are satisfied:

$$\mathbb{E}[\partial_\theta l(W; \theta_0, \beta_0)] = 0, \quad \mathbb{E}[\partial_\beta l(W; \theta_0, \beta_0)] = 0. \quad (34)$$

So in our case the identifying moment condition (27) checks out. However, this score is generally not orthogonal to the nuisance parameter β , namely we generally have

$$\partial_\beta \mathbb{E}[\partial_\theta l(W; \theta_0, \beta_0)] \neq 0,$$

so estimation error in β may affect inference on θ at first order.

To remove this sensitivity, one constructs the orthogonal score

$$\psi(W; \theta, \eta) = \partial_\theta \ell(W; \theta, \beta) - \mu \partial_\beta \ell(W; \theta, \beta), \quad (35)$$

where the nuisance parameter vector is extended from just β' to $\eta = (\beta', \text{vec}(\mu)')' \in \mathcal{B} \times \mathbb{R}^{d_\theta d_\beta}$ and the matrix μ is an orthogonalization matrix so that the constructed score (35) becomes orthogonal to perturbations in the nuisance direction. We now unpack a little how to solve for this μ .

The key is to back-engineer: we simply need to find a μ such that the constructed new score (35) is now orthogonal to influence caused by the nuisance β . Taking the partial derivative with respect β and centering around the truth gives:

$$\partial_\beta \psi = \partial_\beta \partial_\theta \ell - \mu \partial_\beta^2 \ell = 0. \quad (36)$$

Let $J_{\theta\beta}$ denotes the matrix of $\partial_{(\theta', \beta')'}$ valuated at (α_0, β_0) , and $J_{\beta\beta}$ defined similarly, we notice they are just submatrices in the Hessian matrix of the objective function:

$$J = \partial_{(\theta', \beta')'} \mathbb{E}[\partial_{(\theta', \beta')'} \ell(W; \theta, \beta)] \Big|_{\theta=\theta_0, \beta=\beta_0} = \mathbb{E}[\partial_{(\theta', \beta')'}^2 \ell(W; \theta, \beta)] \Big|_{\theta=\theta_0, \beta=\beta_0}. \quad (37)$$

To be more transparent, the four symmetric blocks in J are the corresponding submatrices

<i>Observed (W)</i>	<i>Nuisance parameters (η)</i>	<i>Neyman orthogonal score $\psi(W; \theta, \eta)$</i>
(i) <i>Linear regression coefficient</i> Y, D, X	$\eta_{Y,0} = \arg \min_{\eta} \mathbb{E}[(Y - X'\eta)^2],$ $\eta_{D,0} = \arg \min_{\eta} \mathbb{E}[(D - X'\eta)^2]$	$[(Y - X'\eta_Y) - \theta(D - X'\eta_D)](D - X'\eta_D)$
(ii) <i>Partially linear regression coefficient</i> Y, D, X	$\ell_0(X) = \mathbb{E}[Y X],$ $r_0(X) = \mathbb{E}[D X]$	$[(Y - \ell(X)) - \theta(D - r(X))](D - r(X))$
(iii) <i>Linear IV coefficient (no controls)</i> Y, D, Z	$r_0(Z) = \mathbb{E}[D Z]$	$(Y - \theta D) r(Z)$
(iv) <i>Average treatment effect (D binary)</i> Y, D, X	$\ell_0(D, X) = \mathbb{E}[Y D, X],$ $\alpha_0(D, X) = \frac{D}{r_0(X)} - \frac{1-D}{1-r_0(X)}$ for $r_0(X) = \mathbb{E}[D X]$	$\alpha(D, X)(Y - \ell(D, X)) + \ell(1, X) - \ell(0, X) - \theta$
(v) <i>Partially linear regression with excluded instrument</i> Y, D, Z, X	$\ell_0(X) = \mathbb{E}[Y X]$ $r_0(X) = \mathbb{E}[D X],$ $h_0(X) = \mathbb{E}[Z X]$	$[(Y - \ell(X)) - \theta(D - r(X))](Z - h(X))$

Table 2: Common orthogonal scores in semiparametric inference.

in the form below

$$J = \begin{pmatrix} \partial_{\theta} \mathbb{E}[\partial_{\theta} \ell] & \partial_{\beta} \mathbb{E}[\partial_{\theta} \ell] \\ \partial_{\theta} \mathbb{E}[\partial_{\beta} \ell] & \partial_{\beta} \mathbb{E}[\partial_{\beta} \ell] \end{pmatrix}_{(\theta, \beta) = (\theta_0, \beta_0)} = \begin{pmatrix} J_{\theta\theta} & J_{\theta\beta} \\ J_{\beta\theta} & J_{\beta\beta} \end{pmatrix}.$$

Lastly, (36) implies the orthogonality condition is reduced to solving

$$J_{\theta\beta} - \mu J_{\beta\beta} = 0,$$

which yields the solution

$$\mu = J_{\theta\beta} J_{\beta\beta}^{-1},$$

assuming $J_{\beta\beta}^{-1}$ is full-rank. If this construction using the log-likelihood feels still too technical, Appendix B works through the above procedure in the high dimensional regression setting, where the familiar linear model specification can help with understanding the mathematics.

Similar procedure extends immediately beyond the case of a criterion function. For instance, Chernozhukov et al. (2018) also discusses the construction for GMM settings where the moment condition is more general $\mathbb{E}[m(W, \theta_0, \beta_0)] = 0$, but we omit its derivation due to complex matrix algebra.

4.3 Illustration via Estimation of ATE

Two takeaways from the exercises in Section 4.2 are in order. First, Neyman Orthogonality mechanically ensures that the regularization bias does not propagate in the process of learning the target parameter in first order. Second, for any original, not necessarily orthogonal score φ , we can construct a new orthogonal score by extending the nuisance parameter $\beta \in \mathcal{B}$ to $\eta \in \mathcal{T}$, where $\mathcal{T}$ is usually a superset of $\mathcal{B}$. The second point seems a little counterintuitive, as we are introducing more nuisance elements into estimation. This section aims to elucidate this idea by working through the case of ATE, which is one of the most common causal targets in social scientific inquiry. At the end of this chapter, we hope the readers will see the process of building up to an orthogonal score is the same as finding a “doubly robust” estimator. The extra nuisance parameters are brought in exactly to offset potential errors in learning existing nuisance parameters and ensure such bias does not propagate in the learning of the causal target.

Most generally, the idea is that given $\varphi(W; \theta, \beta)$, we can almost always find an adjustment $\phi(W; \theta, \eta)$ such that

$$\psi(W; \theta, \eta) = \varphi(W; \theta, \beta) + \phi(W; \theta, \eta),$$

as long as ϕ satisfy some nice conditions (Chernozhukov et al., 2018). Table 2 collects the constructed Neyman orthogonal scores for other common causal targets.

For our example, we focus on the ATE, identifiable via

$$\alpha := \text{ATE} = \mathbb{E}[\mathbb{E}[Y(1) - Y(0)] \mid X],$$

where we assume the treatment D is binary and X are given as controls.

Regression Adjustment Estimator. Since we are given controls, one standard approach is to estimate the ATE via regression adjustment:

$$\widehat{\text{ATE}}_{\text{reg}} = \frac{1}{n} \sum_{i=1}^n (\hat{g}_1(x_i) - \hat{g}_0(x_i)), \quad (38)$$

where $g_1(X) = Y(1)$ and $g_0(X) = Y(0)$ represent the true regression equation when the treatment is turned on and off, respectively. Is this a good estimator? It will be consistent fairly generally as long as regression functions flexibly estimated, but it will NOT be generally asymptotically normal unless estimators for $g_1(\cdot)$ and $g_0(\cdot)$ are correctly specified parametric models.

For instance, If one specifies linear models $g_0(x) = x'\beta_0$ and $g_1(x) = x'\beta_1$, then $\widehat{\text{ATE}}_{\text{reg}}$ is equivalent to coefficient on D from regression of Y on D , X , and $D(X - \bar{X})$. The usual OLS

standard errors ignore sampling uncertainty from using $\bar{X}$ in place of $\mathbb{E}[X]$. In another word, it is too strong to assume $\bar{X} \xrightarrow{p} \mathbb{E}[X]$.

What's more, writing in the framework of score function ψ , we are trying to find solution to the moment condition

$$\mathbb{E}[\psi_{\text{reg}}(W; \alpha_0, h_0)] = \mathbb{E}[g_1(X) - g_0(X) - \alpha] = 0, \quad \psi_{\text{reg}}(W; \alpha_0, h_0) = g_1(X) - g_0(X) - \alpha, \quad (39)$$

where $h = (g_1(\cdot), g_0(\cdot))$ collects the nuisance (i.e., the regression functions).

Treating ∂_h as a directional derivative, one can quickly check the regression adjustment score (39) is not orthogonal:

$$\frac{\partial \mathbb{E}[g_1(X) - g_0(X) - \alpha]}{\partial g_1} = 1 \neq 0, \quad \frac{\partial \mathbb{E}[g_1(X) - g_0(X) - \alpha]}{\partial g_0} = -1 \neq 0 \quad (40)$$

Propensity Score Weighted Estimator Another commonly adjustment is via the propensity score, defined as

$$e(X) = \mathbb{P}(D = 1 \mid X), \quad (41)$$

which is widely used due to its balancing property and being sufficient for unconfoundedness (see Appendix A for more details). The Propensity Score Weighting Estimator is

$$\widehat{\text{ATE}}_{\text{prop}} = \frac{1}{n} \sum_{i=1}^n \left[\frac{D_i Y_i}{\widehat{e}(x_i)} - \frac{(1 - D_i) Y_i}{1 - \widehat{e}(x_i)} \right], \quad (42)$$

where $\widehat{e}(x_i)$ is an estimator of the propensity score, typically a logit model $\Lambda(X'\gamma)$ if estimated parametrically. Similar to the regression adjustment estimator, $\widehat{\text{ATE}}_{\text{prop}}$ will generally not be asymptotically normal unless $e(\cdot)$ correctly specified parametric model (or known).

We can define the moment condition (27) and write down $\psi_{\text{prop}}(W; \alpha_0, h_0)$ similar to (39), with $h = e(X)$. One can also try taking the partial derivatives $\partial_h \mathbb{E}[\psi_{\text{prop}}(W; \alpha_0, h_0)]$ and verify ψ_{prop} does not satisfy the orthogonality condition. For a full proof for this statement using pathwise derivative, we refer readers to Appendix A.

Combining Regression Adjustment and Propensity Score Weighting: doubly robust estimator As foreshadowed above, neither (38) nor (3) is Neyman orthogonal. We next show that combining them together orthogonalize our score and naturally yields the “doubly robust” estimator, which was first coined by [Robins and Rotnitzky \(1995\)](#) and highly

recommended in the causal inference literature. We define:

$$\widehat{\text{ATE}}_{\text{robust}} = \frac{1}{n} \sum_{i=1}^n \left[\frac{D_i(Y_i - \hat{g}_1(x_i))}{\hat{e}(x_i)} - \frac{(1 - D_i)(Y_i - \hat{g}_0(x_i))}{1 - \hat{e}(x_i)} + \hat{g}_1(x_i) - \hat{g}_0(x_i) \right]. \quad (43)$$

The orthogonalized score is now:

$$\psi_{dr}(W_i, \alpha, h) = \frac{D_i(Y_i - g_1(X_i))}{e(X_i)} - \frac{(1 - D_i)(Y_i - g_0(X_i))}{1 - e(X_i)} + g_1(X_i) - g_0(X_i) - \alpha. \quad (44)$$

Robins and Rotnitzky (1995) showed that (43) will be consistent and $\sqrt{n}$ -asymptotically normal fairly generally as long as regression functions and propensity score flexibly estimated. One can estimate asymptotic variance as using a plug-in estimator:

$$\hat{V}_r = \frac{1}{n} \sum_{i=1}^n (\hat{\psi}_i - \widehat{\text{ATE}}_{\text{robust}})^2. \quad (45)$$

The estimator remains valid as long as either the propensity score OR regression model is correctly specified (in parametric context), and hence the name “doubly robust”.

Orthogonality property of Doubly Robust Estimator of ATE The moment condition for $\widehat{\text{ATE}}_{\text{robust}}$ has an orthogonality property in that

$$\partial_h \mathbb{E}[\psi_r(W_i, \alpha, h)] \Big|_{\alpha_0, h_0} = 0, \quad (46)$$

where α_0 and h_0 denote the population values. See Appendix A for a formal proof of this statement using pathwise derivatives.

4.4 Heuristic Illustration Using Linear Models

To conclude this chapter, we play with another toy example using linear model specification for all nuisance parameters in the doubly robust estimator to demonstrate it does enjoy desirable asymptotic properties. Namely, it is $\sqrt{n}$ consistent, asymptotically normal, and the asymptotic variance can be consistently estimated. Suppose we model:

$$g_1(X_i) = X_i' \hat{\beta}_1, \quad g_0(X_i) = X_i' \hat{\beta}_0, \quad e(X_i) = X_i' \hat{\gamma}, \quad (47)$$

where by the usual argument of OLS estimator these term enjoy CLT:

$$\sqrt{n}(\hat{\beta}_1 - \beta_1) \xrightarrow{d} N(0, V_1), \quad \sqrt{n}(\hat{\beta}_0 - \beta_0) \xrightarrow{d} N(0, V_0), \quad \sqrt{n}(\hat{\gamma} - \gamma) \xrightarrow{d} N(0, V_\gamma). \quad (48)$$

Then take the Taylor expansion of (43) around the truth $(e(\cdot), g_1(\cdot), g_0(\cdot))$:

$$\widehat{\text{ATE}}_{\text{robust}} = \frac{1}{n} \sum_{i=1}^n \left[\frac{D_i(Y_i - \hat{g}_1(X_i))}{\hat{e}(X_i)} - \frac{(1 - D_i)(Y_i - \hat{g}_0(X_i))}{1 - \hat{e}(X_i)} + \hat{g}_1(X_i) - \hat{g}_0(X_i) \right] \quad (49)$$

$$= \frac{1}{n} \sum_{i=1}^n \left[\frac{D_i(Y_i - g_1(X_i))}{e(X_i)} - \frac{(1 - D_i)(Y_i - g_0(X_i))}{1 - e(X_i)} + g_1(X_i) - g_0(X_i) \right] \quad (50)$$

$$+ \underbrace{\left(\frac{1}{n} \sum_{i=1}^n \left[-\frac{D_i}{e(X_i)} + 1 \right] X_i' \right)}_{a_{1,n}} (\hat{\beta}_1 - \beta_1) \quad (51)$$

$$+ \underbrace{\left(\frac{1}{n} \sum_{i=1}^n \left[\frac{1 - D_i}{1 - e(X_i)} - 1 \right] X_i' \right)}_{a_{2,n}} (\hat{\beta}_0 - \beta_0) \quad (52)$$

$$+ \underbrace{\left(\frac{1}{n} \sum_{i=1}^n \left[-\frac{D_i(Y_i - g_1(X_i))}{e(X_i)^2} - \frac{(1 - D_i)(Y_i - g_0(X_i))}{(1 - e(X_i))^2} \right] X_i' \right)}_{a_{3,n}} (\hat{\gamma} - \gamma) + \text{2nd order.} \quad (53)$$

Equivalently,

$$\widehat{\text{ATE}}_{\text{robust}} = \frac{1}{n} \sum_{i=1}^n (\psi_r(W_i, \alpha_0, h_0) + \alpha_0) + a_{1,n}(\hat{\beta}_1 - \beta_1) + a_{2,n}(\hat{\beta}_0 - \beta_0) + a_{3,n}(\hat{\gamma} - \gamma) + \text{2nd order.} \quad (54)$$

Notice $a_{1,n}(\hat{\beta}_1 - \beta_1)$, $a_{2,n}(\hat{\beta}_0 - \beta_0)$, $a_{3,n}(\hat{\gamma} - \gamma)$ are all $O_p(n^{-1/2}) \cdot O_p(n^{-1/2})$, so they vanish asymptotically when scaled by $\sqrt{n}$. Thus, the doubly robust estimator attains a CLT-flavor result:

$$\sqrt{n}(\widehat{\text{ATE}}_{\text{robust}} - \text{ATE}) \xrightarrow{d} N(0, V_r), \quad (55)$$

where we can estimate asymptotic variance with

$$\hat{V}_r = \frac{1}{n} \sum_{i=1}^n (\hat{\psi}_i - \widehat{\text{ATE}}_{\text{robust}})^2. \quad (56)$$

Remark 1. A final remark is that the build-up towards the doubly robust estimator closely mirrors the process of orthogonalization described in Section 4.2.2. Technically, one can start with a different non-orthogonal score $\varphi(W; \theta, \beta)$ and eventually by adding adjustment and nuisance parameters attain some sort of “doubly robust” specifications.

Estimator	Definition	Components
Regression adjustment	$\widehat{\text{ATE}}_{\text{reg}} = \frac{1}{n} \sum_{i=1}^n (\hat{g}_1(x_i) - \hat{g}_0(x_i))$	• $\hat{g}_1(x)$: estimated regression function for Y_1 • $\hat{g}_0(x)$: estimated regression function for Y_0 • x_i: covariates for unit i
Propensity score weighting	$\widehat{\text{ATE}}_{\text{prop}} = \frac{1}{n} \sum_{i=1}^n \left[\frac{D_i Y_i}{\hat{e}(x_i)} - \frac{(1 - D_i) Y_i}{1 - \hat{e}(x_i)} \right]$	• D_i: treatment indicator • Y_i: observed outcome • $\hat{e}(x)$: estimated propensity score, where $e(x) = \mathbb{P}(D = 1 \mid X = x)$ • x_i: covariates for unit i
Doubly robust	$\widehat{\text{ATE}}_{\text{robust}} = \frac{1}{n} \sum_{i=1}^n \left[\frac{D_i (Y_i - \hat{g}_1(x_i))}{\hat{e}(x_i)} - \frac{(1 - D_i) (Y_i - \hat{g}_0(x_i))}{1 - \hat{e}(x_i)} + \hat{g}_1(x_i) - \hat{g}_0(x_i) \right]$	• D_i: treatment indicator • Y_i: observed outcome • $\hat{g}_1(x)$: estimated regression function for Y_1 • $\hat{g}_0(x)$: estimated regression function for Y_0 • $\hat{e}(x)$: estimated propensity score, where $e(x) = \mathbb{P}(D = 1 \mid X = x)$ • x_i: covariates for unit i

Table 3: Three estimators for the average treatment effect (ATE).

References

- Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., and Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters.
- Kennedy, E. H. (2024). Semiparametric doubly robust targeted double machine learning: a review. *Handbook of statistical methods for precision medicine*, pages 207–236.
- Robins, J. M. and Rotnitzky, A. (1995). Semiparametric efficiency in multivariate regression models with missing data. *Journal of the American Statistical Association*, 90(429):122–129.

Appendix

A Proofs of Important Results

Proof of Theorem 1

Proof: Denote:

$$f(Z \mid \theta) = f(z_1, \dots, z_n \mid \theta).$$

Notice, by definition of a valid density function, we have

$$\int f(Z \mid \theta) dZ = 1.$$

Differentiate both sides with respect to vector θ , under regularity conditions, we have

$$\nabla_{\theta} \int f(Z \mid \theta) dZ = \int \nabla_{\theta} f(Z \mid \theta) dZ = \int \frac{\nabla_{\theta} f(Z \mid \theta)}{f(Z \mid \theta)} f(Z \mid \theta) dZ \quad (57)$$

$$= \int \nabla_{\theta} \log f(Z \mid \theta) f(Z \mid \theta) dZ = \mathbb{E}[\nabla_{\theta} \log f(Z \mid \theta)] \quad (58)$$

$$= \mathbb{E}[\nabla_{\theta} \ell_n(\theta)] = 0. \quad (59)$$

■

Proof of Theorem 2

Proof: Combining (58) and (59), we have

$$\int \nabla_{\theta} \log f(Z \mid \theta) f(Z \mid \theta) dZ = 0.$$

Differentiating both sides again with respect to θ , we obtain:

$$\int \nabla_{\theta}^2 \log f(Z | \theta) f(Z | \theta) dZ + \int \nabla_{\theta} \log f(Z | \theta) \nabla'_{\theta} \log f(Z | \theta) f(Z | \theta) dZ = 0, \quad (60)$$

which implies

$$\mathbb{E}[\nabla_{\theta}^2 \ell_n(\theta)] + \mathbb{E}[\nabla_{\theta} \ell_n(\theta) \nabla'_{\theta} \ell(\theta)] = 0 \quad (61)$$

$$\implies -\mathbb{E}[\nabla_{\theta}^2 \ell_n(\theta)] = \text{Var}[\ell_n(\theta)] \quad (62)$$

■

Proof of Pathwise Differentiability (22)

Proof: Under the regularity condition, let (P_{ε}, P) be such that the von Mises expansion holds:

$$\psi(P_{\varepsilon}) - \psi(P) = \int \varphi(z; P_{\varepsilon}) d(P_{\varepsilon} - P)(z) + R_2(P_{\varepsilon}, P).$$

Differentiate both sides with respect to ε and evaluate at $\varepsilon = 0$. The remainder term satisfies

$$\left. \frac{\partial}{\partial \varepsilon} R_2(P_{\varepsilon}, P) \right|_{\varepsilon=0} = 0,$$

so

$$\left. \frac{\partial}{\partial \varepsilon} \psi(P_{\varepsilon}) \right|_{\varepsilon=0} = \left. \frac{\partial}{\partial \varepsilon} \int \varphi(z; P_{\varepsilon}) d(P_{\varepsilon} - P)(z) \right|_{\varepsilon=0}.$$

Since

$$\left. \frac{\partial}{\partial \varepsilon} (P_{\varepsilon} - P) \right|_{\varepsilon=0} = \left. \frac{\partial}{\partial \varepsilon} P_{\varepsilon} \right|_{\varepsilon=0},$$

and by the definition of the submodel,

$$\left. \frac{\partial}{\partial \varepsilon} P_{\varepsilon} \right|_{\varepsilon=0} = h(z) dP(z),$$

we obtain

$$\left. \frac{\partial}{\partial \varepsilon} \psi(P_{\varepsilon}) \right|_{\varepsilon=0} = \int \varphi(z; P) h(z) dP(z).$$

Because h is the score function of the submodel at $\varepsilon = 0$, this can be written as

$$\left. \frac{\partial}{\partial \varepsilon} \psi(P_{\varepsilon}) \right|_{\varepsilon=0} = \int \varphi(z; P) s_{\varepsilon}(z) dP(z),$$

which is the desired result. ■

Balancing Properties of Propensity Score

Theorem 4. *Unconfoundedness of treatment implies unconfoundedness of propensity score and vice versa. Formally:*

$$D \perp\!\!\!\perp X \mid e(X) \iff \mathbb{P}(D = 1 \mid X, e(X)) = \mathbb{P}(D = 1 \mid e(X)). \quad (63)$$

Proof: (i)

$$\mathbb{P}(D = 1 \mid X, e(X)) = \mathbb{P}(D = 1 \mid X) = e(X).$$

(ii)

$$\mathbb{P}(D = 1 \mid e(X)) = \mathbb{E}[\mathbb{1}\{D = 1\} \mid e(X)] = \mathbb{E}[\mathbb{E}[D \mid X, e(X)] \mid e(X)] = \mathbb{E}[e(X) \mid e(X)] = e(X). \quad \blacksquare$$

Unconfoundedness Given the Propensity Score If treatment assignment is unconfounded, then treatment assignment is unconfounded given the propensity score:

$$D \perp (Y_1, Y_0) \mid e(X). \quad (64)$$

Proof:

$$\mathbb{P}(D = 1 \mid Y_0, Y_1, e(X)) = \mathbb{E}[D \mid Y_0, Y_1, e(X)] \quad (65)$$

$$= \mathbb{E}[\mathbb{E}[D \mid Y_0, Y_1, e(X), X] \mid Y_0, Y_1, e(X)]. \quad (66)$$

By unconfoundedness and the balancing property,

$$\mathbb{E}[D \mid Y_0, Y_1, e(X), X] = \mathbb{E}[D \mid e(X)].$$

Plugging in gives

$$\mathbb{E}[\mathbb{E}[D \mid e(X)] \mid Y_0, Y_1, e(X)] = \mathbb{E}[D \mid e(X)] = \mathbb{P}(D = 1 \mid e(X)).$$

So conditioning on the propensity score is sufficient. \blacksquare

Propensity Score Weighted Estimator for ATE is not Neyman Orthogonal

Proof: Let

$$\psi(W; \alpha, e) = \frac{DY}{e(X)} - \frac{(1-D)Y}{1-e(X)} - \alpha,$$

where α denotes the average treatment effect. The corresponding population moment is

$$M(\alpha, e) = \mathbb{E}[\psi(W; \alpha, e)].$$

At the true value α_0 , the estimator would be Neyman orthogonal with respect to the nuisance e if the pathwise derivative of $M(\alpha_0, e)$ in any direction h were zero.

Consider the perturbed propensity score

$$e_t(x) = e(x) + th(x),$$

where h is an arbitrary bounded direction and t is small enough so that $e_t(x) \in (0, 1)$. Then

$$M(\alpha_0, e_t) = \mathbb{E}\left[\frac{DY}{e_t(X)} - \frac{(1-D)Y}{1-e_t(X)} - \alpha_0\right].$$

Differentiating with respect to t at $t = 0$ gives

$$\left.\frac{d}{dt}M(\alpha_0, e_t)\right|_{t=0} = \mathbb{E}\left[-DY \frac{h(X)}{e(X)^2} - (1-D)Y \frac{h(X)}{(1-e(X))^2}\right].$$

Equivalently,

$$\left.\frac{d}{dt}M(\alpha_0, e_t)\right|_{t=0} = -\mathbb{E}\left[h(X) \left\{ \frac{DY}{e(X)^2} + \frac{(1-D)Y}{(1-e(X))^2} \right\}\right].$$

Since this derivative is not identically zero for arbitrary perturbation directions h , the score is not Neyman orthogonal with respect to the nuisance e . Thus, the inverse-probability-weighted ATE score is not orthogonal in the nuisance direction e . ■

Doubly robust estimator of ATE obeys Neyman orthogonality

Proof: Consider the score

$$\psi_{dr}(W; \alpha, \eta) = \frac{D(Y - g_1(X))}{e(X)} - \frac{(1-D)(Y - g_0(X))}{1-e(X)} + g_1(X) - g_0(X) - \alpha,$$

where $\eta = (e, g_0, g_1)$. Let

$$M(\alpha, \eta) = \mathbb{E}[\psi_{dr}(W; \alpha, \eta)].$$

To verify Neyman orthogonality, we must show that the pathwise derivative of $M(\alpha_0, \eta)$ with respect to each nuisance component vanishes at the truth $\eta_0 = (e_0, g_{0,0}, g_{1,0})$.

We first perturb the propensity score by $e_t(x) = e_0(x) + th_e(x)$. Then

$$\left.\frac{d}{dt}M(\alpha_0, e_t, g_{0,0}, g_{1,0})\right|_{t=0} = \mathbb{E}\left[-\frac{D(Y - g_{1,0}(X))}{e_0(X)^2}h_e(X) + \frac{(1-D)(Y - g_{0,0}(X))}{(1-e_0(X))^2}h_e(X)\right].$$

Using the defining properties of the regression functions,

$$\mathbb{E}[Y - g_{1,0}(X) \mid X, D = 1] = 0, \quad \mathbb{E}[Y - g_{0,0}(X) \mid X, D = 0] = 0,$$

we obtain

$$\mathbb{E}\left[\frac{D(Y - g_{1,0}(X))}{e_0(X)^2} h_e(X)\right] = 0, \quad \mathbb{E}\left[\frac{(1 - D)(Y - g_{0,0}(X))}{(1 - e_0(X))^2} h_e(X)\right] = 0.$$

Hence

$$\left. \frac{d}{dt} M(\alpha_0, e_t, g_{0,0}, g_{1,0}) \right|_{t=0} = 0.$$

Next perturb g_1 by $g_{1,t}(x) = g_{1,0}(x) + th_1(x)$. Then

$$M(\alpha_0, e_0, g_{0,0}, g_{1,t}) = \mathbb{E}\left[\frac{D(Y - g_{1,t}(X))}{e_0(X)} - \frac{(1 - D)(Y - g_{0,0}(X))}{1 - e_0(X)} + g_{1,t}(X) - g_{0,0}(X) - \alpha_0\right],$$

so

$$\left. \frac{d}{dt} M(\alpha_0, e_0, g_{0,0}, g_{1,t}) \right|_{t=0} = \mathbb{E}\left[-\frac{D}{e_0(X)} h_1(X) + h_1(X)\right].$$

Since $\mathbb{E}[D \mid X] = e_0(X)$, this becomes

$$\mathbb{E}\left[\left(1 - \frac{D}{e_0(X)}\right) h_1(X)\right] = \mathbb{E}\left[\underbrace{\mathbb{E}\left[1 - \frac{D}{e_0(X)} \mid X\right]}_0 h_1(X)\right] = 0.$$

Thus, the derivative with respect to g_1 vanishes. The same argument shows that perturbing g_0 by $g_{0,t}(x) = g_{0,0}(x) + th_0(x)$ yields

$$\left. \frac{d}{dt} M(\alpha_0, e_0, g_{0,t}, g_{1,0}) \right|_{t=0} = 0.$$

Therefore the score ψ_{dr} satisfies the Neyman orthogonality condition:

$$\left. \frac{\partial}{\partial t} \mathbb{E}[\psi_{dr}(W; \alpha_0, \eta_t)] \right|_{t=0} = 0$$

for every regular perturbation direction of the nuisance parameter $\eta = (e, g_0, g_1)$. ■

B Additional Example

Construction of orthogonal score using likelihood in high dimensional regression

As an illustration of the construction in Section 4.2.2, consider the following linear predictive

model:

$$\boxed{\begin{aligned} Y &= D\theta_0 + X'\beta_0 + U, & \mathbb{E}[U \mid X, D] &= 0, \\ D &= X'\gamma_0 + V, & \mathbb{E}[V \mid X] &= 0. \end{aligned}}$$

For simplicity, assume that θ_0 is scalar, while $\beta_0 \in \mathbb{R}^{d_\beta}$. The first equation is the main predictive model, and the second equation is introduced only to construct the Neyman orthogonal score.

The pair (θ_0, β_0) solves the least-squares problem

$$\max_{\theta \in \Theta, \beta \in \mathcal{B}} \mathbb{E}[\ell(W; \theta, \beta)], \quad \ell(W; \theta, \beta) = -\frac{(Y - D\theta - X'\beta)^2}{2},$$

where $W = (Y, D, X')'$, $\theta \in \Theta = \mathbb{R}$, and $\beta \in \mathcal{B} = \mathbb{R}^{d_\beta}$.

The corresponding score functions are

$$\partial_\theta \ell(W; \theta, \beta) = (Y - D\theta - X'\beta)D, \quad \partial_\beta \ell(W; \theta, \beta) = (Y - D\theta - X'\beta)X.$$

Differentiating once more gives the Hessian blocks:

$$\begin{aligned} \partial_{\theta\theta} \ell(W; \theta, \beta) &= -D^2, & \partial_{\theta\beta} \ell(W; \theta, \beta) &= -DX', \\ \partial_{\beta\theta} \ell(W; \theta, \beta) &= -XD, & \partial_{\beta\beta} \ell(W; \theta, \beta) &= -XX'. \end{aligned}$$

Hence, the matrix of expected second derivatives is

$$J = \mathbb{E}[\nabla_{(\theta, \beta)}^2 \ell(W; \theta, \beta)]_{(\theta, \beta) = (\theta_0, \beta_0)} = \begin{pmatrix} J_{\theta\theta} & J_{\theta\beta} \\ J_{\beta\theta} & J_{\beta\beta} \end{pmatrix},$$

with

$$J_{\theta\theta} = -\mathbb{E}[D^2], \quad J_{\theta\beta} = -\mathbb{E}[DX'], \quad J_{\beta\theta} = -\mathbb{E}[XD], \quad J_{\beta\beta} = -\mathbb{E}[XX'].$$

The original score for θ is

$$\varphi(W; \theta, \beta) = \partial_\theta \ell(W; \theta, \beta) = (Y - D\theta - X'\beta)D,$$

but this score is not generally orthogonal to the nuisance parameter β . To enforce orthogonality,

define the Neyman orthogonal score

$$\psi(W; \theta, \eta) = (Y - D\theta - X'\beta)(D - \mu X), \quad \eta = (\beta', \text{vec}(\mu)')'.$$

The orthogonalization matrix μ is chosen so that

$$J_{\theta\beta} - \mu J_{\beta\beta} = 0.$$

Using the explicit formulas above, this becomes

$$-\mathbb{E}[DX'] - \mu(-\mathbb{E}[XX']) = 0,$$

or equivalently,

$$\mu \mathbb{E}[XX'] = \mathbb{E}[DX'].$$

If $\mathbb{E}[XX']$ is invertible, then

$$\mu_0 = \mathbb{E}[DX'] (\mathbb{E}[XX'])^{-1} = \gamma'_0,$$

where the last equality follows from the linear projection equation $D = X'\gamma_0 + V$ with $\mathbb{E}[V | X] = 0$.

Hence, the orthogonal score at the truth is

$$\psi(W; \theta_0, \eta_0) = U(D - \mu_0 X),$$

which is orthogonal to first-order perturbations in the nuisance direction.